

Research Statement

My primary mathematical interest lies in algebraic geometry, with a current focus on the syzygies of linear series. I am fascinated by how geometric shapes are connected with abstract algebra, especially commutative algebra in algebraic geometry. I hope to further study topics and problems in algebraic geometry, which is the fundamental reason why I want to pursue a doctorate in mathematics. I also find great joy in discovering problems and doing creative work in my research, which motivates me to continue on my academic path. My long-term goal is to contribute original research to the field and to mentor students who share my passion for mathematics.

I first came into contact with algebraic geometry in my undergraduate thesis “Minkowski Sum of Lattice Points in Lattice Polytopes” at Sun Yat-sen University. This project introduced me to toric varieties, and I was attracted by how they transform complicated geometric structures into accessible combinatorial data derived from polytopes and fans, which serves as a dictionary between algebraic geometry and combinatorics. My work led to the paper “On covering simplices by dilations in dimensions 3 and 4” on arXiv. Through this process, I learned the fundamentals of algebraic geometry and discovered great joy in exploring problems, which felt like “peeling an onion,” where each layer reveals a new level of understanding. This research experience sparked my enduring interest in algebraic geometry.

Because of my outstanding academic record and overall performance, I was recommended for admission to the master’s program without taking the entrance examination. During my graduate studies, I completed advanced courses such as Riemannian Geometry, Algebraic Topology, and Commutative Algebra with excellent results, which strengthened my foundation in pure mathematics. Furthermore, I gave a series of talks in small reading seminars on Introduction to Toric Varieties and Graded Syzygies at my university. Serving as a speaker allowed me to gain deeper insights through discussions and enhanced my communication skills, particularly in learning how to focus on the core of a problem and convey complex ideas clearly. For my strong academic performance, I received the Graduate Scholarship from my university for three consecutive years.

Beyond coursework, I have also actively participated in various academic activities to communicate with algebraic geometers from all over the world and to appreciate different flavors of algebraic geometry. For instance, I learned about multiplier ideals, V -filtrations, and K -stability theory of singularities during the Algebraic Geometry Summer School held in Shanghai in 2024, as well as geometric aspects of Hodge theory and arithmetic aspects of algebraic geometry in 2025 Summer Research Institute in Algebraic Geometry held at Colorado State University.

My current research focuses on toric geometry, particularly the positivity of linear series and syzygies. The concept of syzygies originated in the 19th century with the work of Sylvester, Cayley, and Hilbert, describing the shape of free resolutions and Betti numbers in commutative algebra. From the 20th century onward, syzygies became a bridge between algebraic geometry

and commutative algebra, connecting the structure of minimal free resolutions to the geometry of embedded projective varieties. In particular, syzygies describes how a variety is embedded, such as its projective normality, generation by quadrics, and higher-order relations.. My research has exposed me to a variety of topics, including GIT theory, stability of syzygy bundles, Klyachko's classification, and resolutions of toric subvarieties. This work has taught me to navigate both abstract theory and concrete computation, and to approach problems from multiple perspectives. More importantly, it has shown me the essence of research: to remain curious amid uncertainty and to learn and discover through trial and error.

In addition, during the first year of my master's program, I explored tropical geometry and wrote a survey on "Curve Counting and Tropical Geometry" for a graduate-level course. In both toric and tropical contexts, I combined theoretical reasoning with extensive computational experiments. Using tools such as Python and Macaulay2, I implemented algorithms to test projective normality, compute syzygies, and verify covering properties of lattice polytopes.

Teaching is also an important part of my development. As a teaching assistant for undergraduate linear algebra courses, I assisted instructors and tutored students, solving student's problem and building a supportive learning environment. I believe that teaching is a crucial way to refine one's own understanding of mathematical concepts and support the next generation of mathematicians.

In conclusion, my academic path has been guided by curiosity about the structures in algebraic geometry and by persistence in tackling challenging problems. In the future, I hope to further broaden my understanding of the field, strengthen my ability to conduct independent research, and contribute to the frontiers of algebraic geometry through original work.